# VehicleGuard AI

## Intelligent Vehicle Theft Detection System

Computer vision that flags suspicious vehicle activity for human review.




Computer Vision • Object Tracking • Explainable Risk Alerts

# Problem & Objective

## The Problem

Vehicle thefts in parking areas are increasing, and CCTV systems generate hours of video, making it difficult to manually spot and verify suspicious activities in real time.

## Our Objective

To build an **AI-based** system that automatically analyses video, understands **person–vehicle interactions**, and raises alerts for potential vehicle thefts with evidence.

## Why It Matters?

### Improves Security

Continuously monitors vehicles and people for unusual behavior.

### Saves Time

Automatically highlights important events from long video recordings.

### Enables Quick Response

Generates real-time alerts with snapshots and short video clips.

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# Proposed Solution & Key Features

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## Our Solution

A computer vision system that analyzes video, tracks vehicles and people, links their activities, and detects suspicious behavior to alert possible vehicle thefts with evidence.



## Key Features

### Vehicle & Person Detection
Detects vehicles and people with instance segmentation.

### Theft Risk Analysis
Identifies suspicious actions (e.g., unauthorized entry, unusual movement).

### Tracking & Association
Gives unique IDs and links people with the vehicle they exit or enter.

### Evidence Recording
Saves alert snapshots, short video clips, and event logs.

### Re-identification & Clothing Analysis
Recognizes returning people using appearance and clothing features.

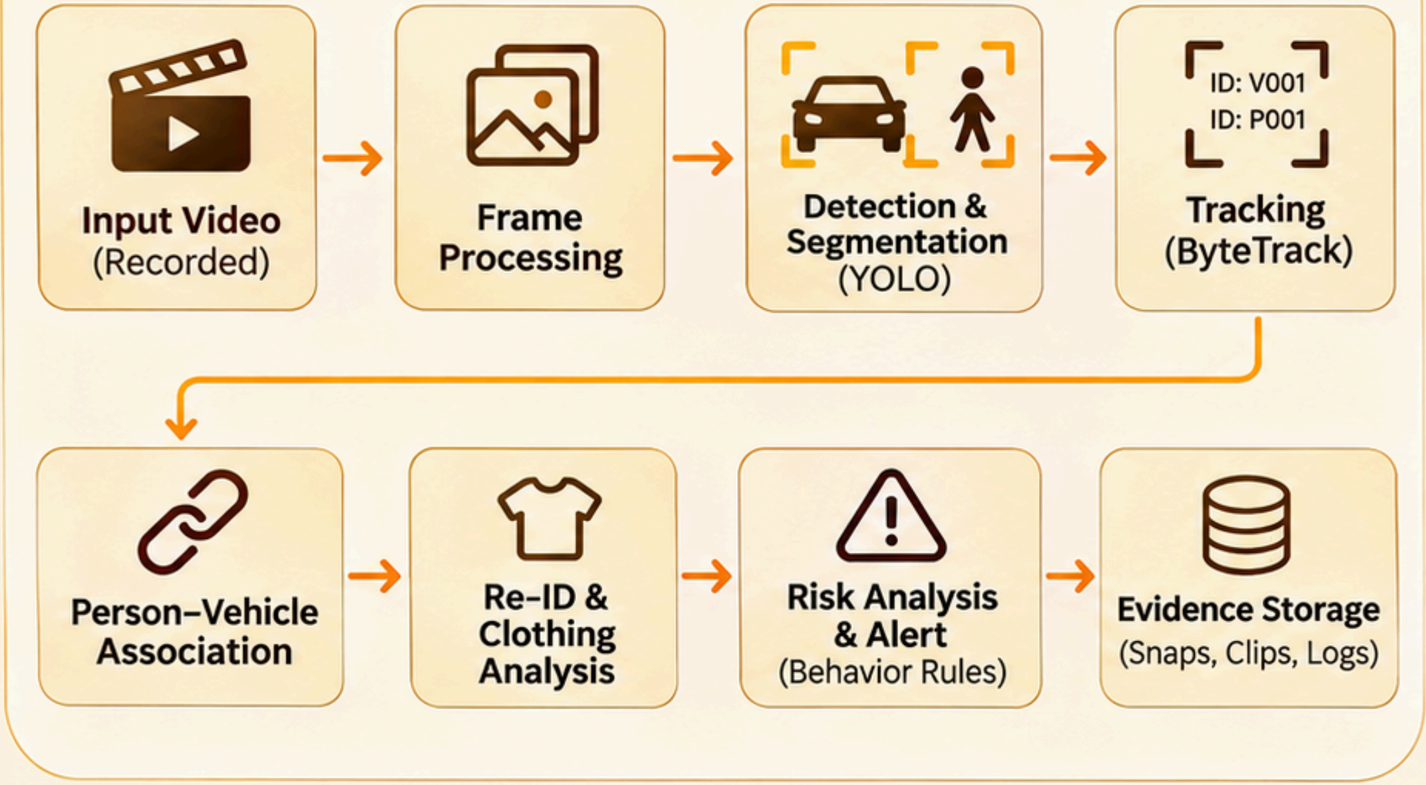
### Parking Zone Monitoring
Monitors vehicle entry, parking, movement, and exit within defined zones.

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# System Architecture & Workflow

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## System Architecture




## Technologies Used

- **Python 3.11** – Core development
- **YOLO (Segmentation)** – Detects & segments vehicles and people
- **ByteTrack** – Multi-object tracking
- **PyTorch + CUDA** – GPU-accelerated inference (RTX 3050)
- **OpenCV** – Video processing and visualization
- **SQLite** – Store alerts, events and associations

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# AI Models & Computer Vision Techniques

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## Models Used

### YOLO (Segmentation)

Detects and segments people and vehicles with instance masks.

### ByteTrack

Tracks detected objects across frames and gives stable IDs (e.g., V001, P001).

### Re-ID & Clothing Analysis

Matches returning people using appearance features and analyzes visible clothing (e.g., shirt colour, type).

## Key Computer Vision Techniques




Input Frame → Object Detection → Instance Segmentation → Tracking & Association

## Risk Analysis (Behavior Rules)

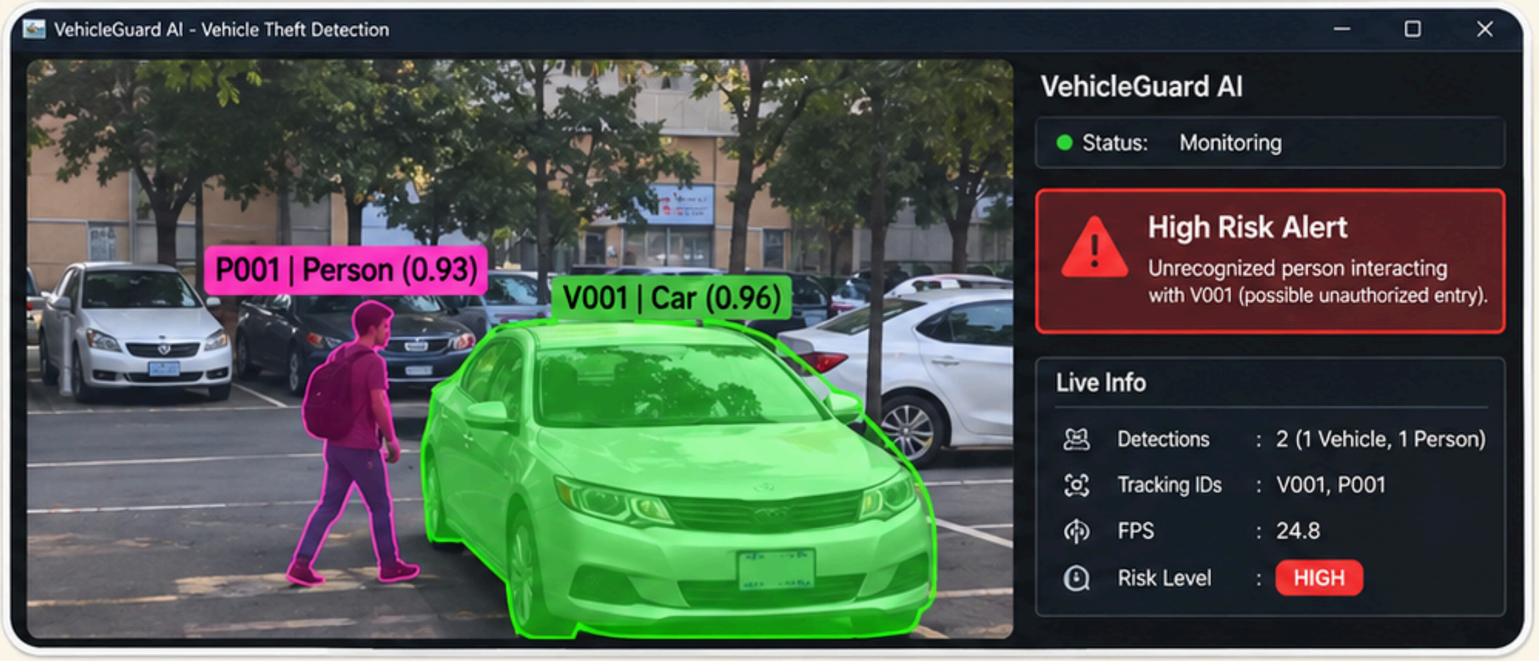
- Unusual or prolonged presence near a vehicle
- Possible unauthorized entry into a vehicle
- Vehicle movement from the parking zone

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# Implementation & Working

## How It Works?

1. **Load Video**
   Select a recorded parking video.
2. **Detect & Segment**
   Identify people and vehicles with masks.
3. **Track & Associate**
   Give unique IDs and link people with the vehicle they interact with.
4. **Analyze Behavior**
   Check for suspicious activities (e.g., unauthorized entry, vehicle movement).
5. **Generate Alert**
   Show risk level and save evidence (snapshot, clip, and event log).




## Output & Evidence

**Alert Snapshot**



**Event Video Clip**



**Event Log (SQLite)**

| Event ID | Vehicle ID | Person ID | Risk |
|---|---|---|---|
| E001 | V001 | P001 | HIGH |

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# Results, Conclusion & Future Scope

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## Results & Outcomes

- Successfully **detects and segments** vehicles and people using YOLO.
- **Tracks multiple objects** with unique IDs (e.g., V001, P001) using ByteTrack.
- Identifies suspicious vehicle interactions and generates **risk alerts** with evidence (snapshots, video clips, and logs).

## Applications



**Parking Areas**
(Colleges, Malls, IT Parks)



**Campus /**
Industrial Security



**Vehicle**
Monitoring & Safety

## Conclusion

**VehicleGuard AI** automates the analysis of parking video, understands person–vehicle interactions, and highlights suspicious activities with supporting evidence, helping in quicker and more effective monitoring.

## Future Scope

- Integrate live CCTV or multi-camera support.
- Improve re-identification and clothing analysis with larger datasets.
- Add an admin dashboard for alert review and vehicle management.
- Explore deployment on cloud/edge devices for real-time monitoring.

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# Thank You!

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